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CLOUD · DATA · OBSERVABILITY · SRE

FIELD HANDBOOK · DAY 5 · SPLUNK POWER USER TRACK

Advanced SPL, Performance & Tuning

The commands that correlate, reshape and forecast — **join**, **transaction**, **xyseries**, **multivalue**, **subsearches**, **timewrap** and the **anomaly-detection** family — plus how to troubleshoot searches and tune Splunk for speed at scale.

join · transaction · subsearch

xyseries · multivalue

anomalydetection · predict

Data Model & Pivot

Troubleshooting

Performance Tuning

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Contents

ORIENTATION

Advanced SPL & the Cost Model	01
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PART A · ADVANCED COMMANDS & FEATURES

1 · join — Correlating Result Sets (& alternatives)	02
2 · lookup — Advanced Enrichment	03
3 · transaction — Grouping Related Events	04
4 · Reshaping: xyseries & untable	05
5 · streamstats & eventstats — Windowed Stats	06
6 · Multivalue Commands	07
7 · Subsearches	08
8 · Time Wrapping (timewrap)	09
9 · Anomaly Detection & Forecasting	10
10 · Data Models & Pivot	11

PART B · USER SEARCH PROBLEMS

11 · Troubleshooting Searches	12
-------------------------------	----

PART C · PERFORMANCE TUNING

12 · Making Splunk Fast — Search Optimization	13
13 · Platform, Indexing & Resource Planning	14

APPENDICES

A · Expensive-Pattern → Fast-Alternative Guide	15
B · Day-5 Quick-Reference Cheat Sheet	16

01 · Advanced SPL & the Cost Model

The advanced commands are powerful — and some are expensive. Using them well means understanding *where a search spends its time*. Every search moves through stages, and the golden rule never changes: **reduce the data as early as possible**, because every later command pays for every event that reaches it.

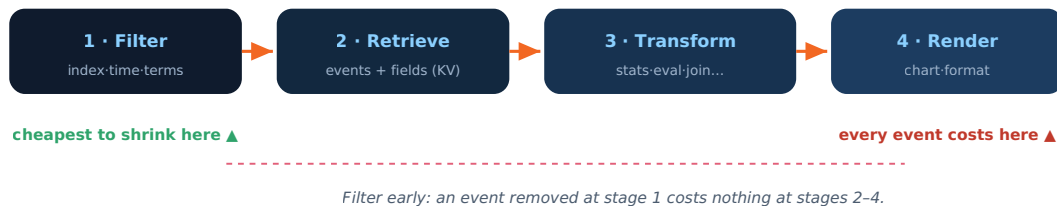


Fig 1.1 — Where a search spends time. Advanced commands live at stage 3; feed them as little as possible.

Streaming vs non-streaming — why it matters here

Class	Examples	Cost behaviour
Streaming	<code>eval</code> , <code>rex</code> , <code>where</code> , <code>fields</code>	Per-event, distributable to indexers — cheap, scales.
Centralised / non-streaming	<code>stats</code> , <code>transaction</code> , <code>join</code> , <code>sort</code>	Run on the search head over the whole set — the expensive stage.

THE THROUGH-LINE FOR DAY 5

Several commands in Part A (`join`, `transaction`, subsearches) are centralised and bounded by limits. They're the right tool sometimes — but Part C shows a faster, safer alternative for most cases. We flag both throughout, on the same Buttercup Games dataset used all week.

PART A

Advanced Commands & Features

Correlate, reshape, expand and forecast. This part is the power-user's toolbox — each command with its best use, its worked example on the dataset, and, where relevant, the faster alternative you should usually reach for first.

02 · join — Correlating Result Sets

`join` merges the current results with the results of a subsearch on one or more shared fields — SQL-style inner/left join. It's intuitive coming from databases, but in Splunk it's often the *wrong* first choice: the subsearch is bounded (default 50,000 rows) and runs centrally, so a big join silently truncates and drags.

Syntax

```
| join type=inner|left clientip [ search index=other | fields clientip ... ]
```

Example — and the better way

```
# join: correlate web sessions with auth events by IP (bounded, slower)
index=web | join clientip [ search index=security | rename src_ip AS clientip | fields clientip
vendor_action ]

# preferred: one pass with stats, no subsearch, no 50k cap
(index=web OR index=security)
| eval ip=coalesce(clientip, src_ip)
| stats count(action) AS web_hits, count(vendor_action) AS auth_events BY ip
```

JOIN PITFALLS

The subsearch is capped (default 50k results, 60s) and finalises silently when it hits the cap — your join then *drops* matches with no error. It also can't stream. For one-to-many or large sets, use `stats` (as above), `lookup`, or `append` + `stats` instead.

WHEN JOIN IS FINE

Small, bounded right side (a few thousand rows) and a genuine need for row-level column merging. Otherwise the `stats ... BY key` pattern is faster, uncapped, and scales.

03 • lookup — Advanced Enrichment

Day 2 covered lookups end-to-end; here we focus on the *advanced* matching modes that make lookups a scalable alternative to `join`. A lookup enriches every event from a table without a subsearch and without a row cap — ideal for one-to-many correlation.

Feature	What it enables
<code>OUTPUTNEW</code>	Write output fields only when currently null — safe enrichment that won't clobber existing values.
Wildcard match	<code>WILDCARD(field)</code> in the definition matches patterns (e.g. URL prefixes).
CIDR match	<code>CIDR(subnet)</code> maps an IP to the subnet row it falls in — network enrichment.
Time-bounded	A time field in the lookup limits matches to a validity window (point-in-time enrichment).
Automatic lookup	Applied to every search of a sourcetype — enrichment "just appears".

Example — CIDR enrichment as a join replacement

```
# map source IPs to network zones via a CIDR lookup – no join, no cap
index=security
| lookup network_zones src_ip OUTPUT zone, owner
| stats count BY zone
```

LOOKUP VS JOIN

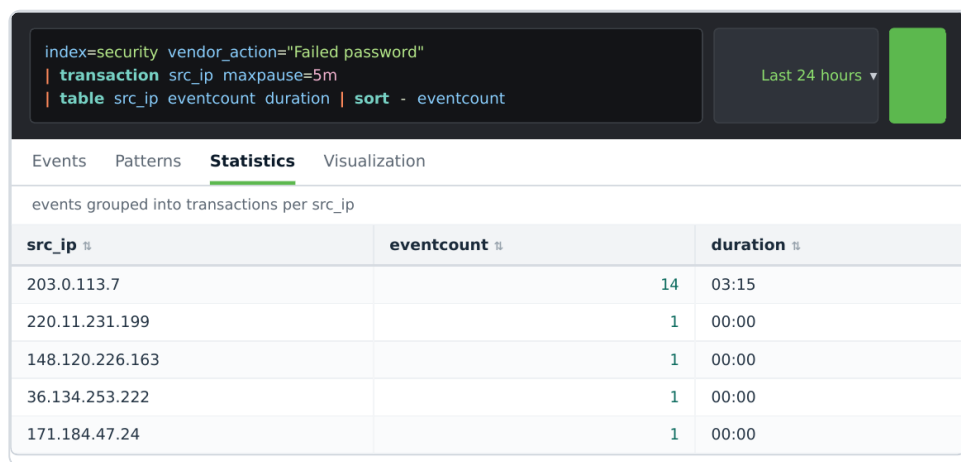
If you're joining just to *add columns from a reference set*, a lookup is almost always the right answer: it streams, has no 50k cap, supises wildcard/CIDR/time matching, and can be made automatic. Reserve `join` for merging two live search result sets.

04 · transaction — Grouping Related Events

`transaction` groups a sequence of events that share field values and fall within time/count bounds into a single multivalued "transaction" event — with handy fields like `duration` and `eventcount`. It's perfect when the *span and order* of a group matter (a session, an attack burst, a checkout flow).

Example — reconstruct attack bursts per source IP

```
index=security vendor_action="Failed password"
| transaction src_ip maxpause=5m
| table src_ip eventcount duration | sort - eventcount
```



The screenshot shows a Splunk search interface. The search bar contains the query: `index=security vendor_action="Failed password" | transaction src_ip maxpause=5m | table src_ip eventcount duration | sort - eventcount`. The results are displayed in a table under the 'Statistics' tab. The table has three columns: `src_ip`, `eventcount`, and `duration`. The data shows that the IP 203.0.113.7 has 14 events with a duration of 03:15, while the other four IPs have 1 event each with a duration of 00:00.

src_ip	eventcount	duration
203.0.113.7	14	03:15
220.11.231.199	1	00:00
148.120.226.163	1	00:00
36.134.253.222	1	00:00
171.184.47.24	1	00:00

Fig 4.1 — The planted attacker 203.0.113.7 becomes one 14-event transaction spanning ~3 minutes.

Option	Bounds a transaction by...
<code>maxpause</code>	Max gap allowed between consecutive events.
<code>maxspan</code>	Max total duration.
<code>startswith / endswith</code>	Events that open/close the group (e.g. login → logout).
<code>maxevents</code>	Cap events per transaction.

TRANSACTION IS EXPENSIVE

It's memory-heavy and non-streaming. When you only need aggregates (count, first/last time, duration), `stats ... by src_ip` with `range(_time)` / `earliest()` / `latest()` is far cheaper. Use `transaction` only when you truly need the grouped events or start/end semantics.

05 • Reshaping: xyseries & untable

Sometimes the numbers are right but the *shape* is wrong for the chart you want. `xyseries` pivots a long, tidy table into a wide matrix (rows × columns × value); `untable` does the reverse. They're the manual controls behind what `chart` does automatically.

Long → wide with xyseries

```
index=web | stats count BY categoryId, action
| xyseries categoryId action count
```

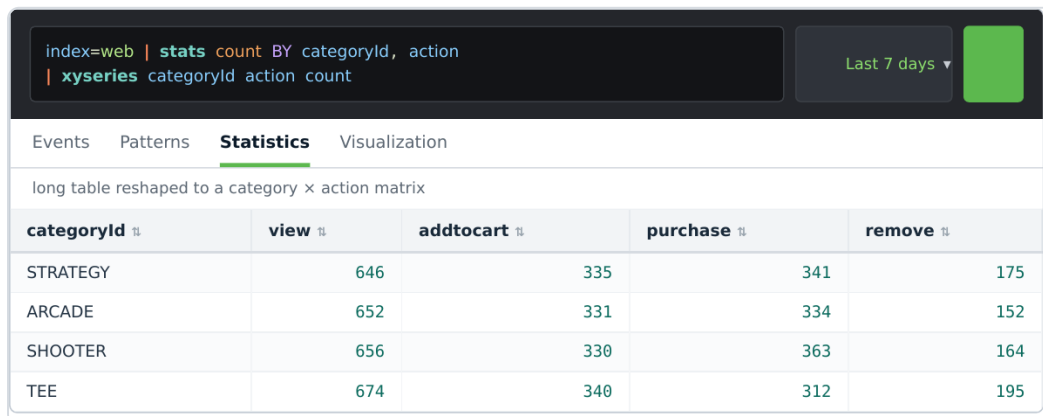


Fig 5.1 — `xyseries <row> <column> <value>` turns three columns into a category × action matrix.

Wide → long with untable

```
# invert a matrix back to row/column/value triples
| untable categoryId action count
```

WHY RESHAPE BY HAND

`chart` pivots for you, but `xyseries` / `untable` give explicit control — useful after custom `stats`, for feeding a specific visualization, or for transposing data before `appendcols` or export.

06 • streamstats & eventstats — Windowed Statistics

Both add aggregate fields *without collapsing* events (unlike `stats`). `eventstats` computes over the **whole** result set and attaches the same value to every row; `streamstats` computes **cumulatively** in order — running totals, moving averages, and time-windowed counts. Day 1 introduced them; here are the advanced controls.

Control (streamstats)	Effect
<code>window=N</code>	Moving window of the last N events.
<code>time_window=5m</code>	Window by time rather than event count.
<code>current=f</code>	Exclude the current event (compare to the running state <i>before</i> it).
<code>reset_on_change=t</code> / <code>reset_before</code>	Restart the accumulation on a condition (per session/day).
<code>global=f ... BY</code>	Independent accumulation per group.

Example — spike detection with a time window

```
index=security vendor_action="Failed password" | sort 0 _time
| streamstats time_window=5m count AS fails_5m BY src_ip
| where fails_5m >= 5
```

EVENTSTATS FOR RATIOS

Use `eventstats sum(price) AS grand_total` then `eval pct = 100*price/grand_total` — each row keeps its detail *and* its share of the whole, in one streaming-friendly pass (no join back to a total).

07 · Multivalue Commands

A single field can hold multiple values (JSON arrays, `values()` results, `rex max_match=0`). The multivalue family lets you build, split, count, filter and explode these fields.

Command / fn	Purpose
<code>mvexpand</code>	Explode a multivalue field into one event per value.
<code>makemv</code> / <code>nomv</code>	Split a delimited string into MV / collapse MV back to a string.
<code>mvcount()</code>	Number of values in the field.
<code>mvindex()</code> / <code>mvfilter()</code> / <code>mvjoin()</code>	Pick, filter, or join values.
<code>split()</code> / <code>mvzip()</code> / <code>mvappend()</code>	Build MV fields in <code>eval</code> .

Example — explode cart items and rank products

```
index=web sourcetype=checkout_json | spath
| eval pid='cart.items{}.productId' | mvexpand pid
| stats count BY pid | sort - count
```

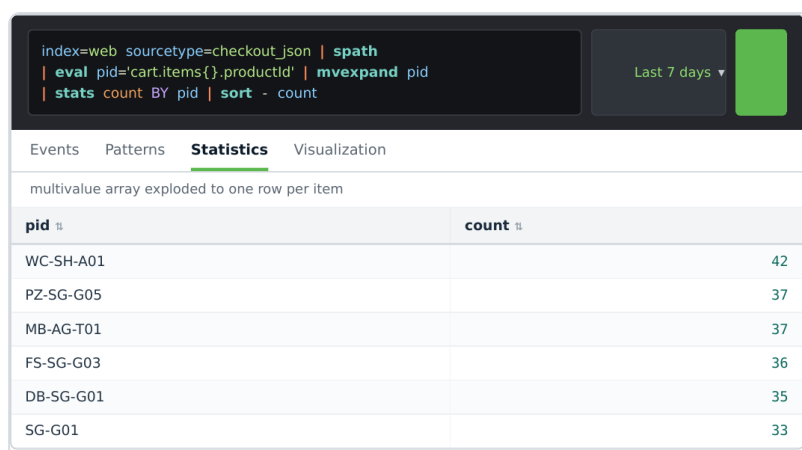


Fig 7.1 — Each basket's `items[]` array explodes to one row per product, then ranks.

MVEXPAND MULTIPLIES EVENTS

Exploding a field with many values multiplies your event count — costly on large sets. Filter first, and prefer MV functions (`mvcount`, `mvfilter`) when you don't actually need one row per value.

08 • Subsearches

A subsearch runs *first*, in square brackets, and its results are substituted into the outer search — typically as a filter term or a set of values. Powerful for "find X, then use it", but bounded and sequential, so it must stay small.

Example — filter the main search by a computed value

```
# outer search filtered by the top revenue category, found by the subsearch
index=web action=purchase
[ search index=web action=purchase | top 1 categoryId | fields categoryId ]
| timechart span=1d count AS orders
```

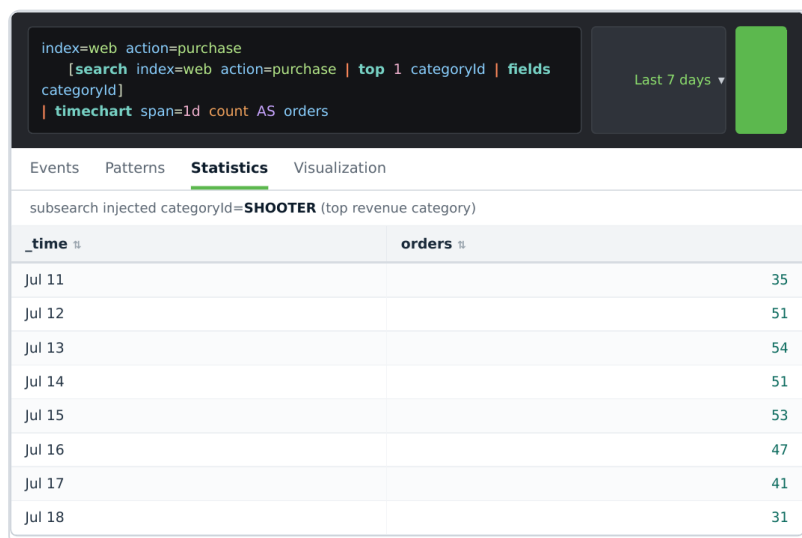


Fig 8.1 — The subsearch injects `categoryId=SHOOTER`; the outer search charts that category's daily orders.

Helper	Use
<code>format</code>	Control how subsearch rows become an outer filter expression.
<code>return</code>	Return specific values/fields compactly from a subsearch.
<code> append [...]</code>	Append subsearch results as extra rows (vs. filtering).

SUBSEARCH LIMITS

Default caps: 10,000 result rows and 60 seconds, then it silently finalises. It also can't stream and runs before the main search. For large correlations prefer `stats` / `lookup`; keep subsearches to small, fast lookups of a handful of values.

09 • Time Wrapping (timewrap)

`timewrap` overlays equal time periods so you can compare like-for-like — this week vs last week, today vs yesterday. It takes the output of a `timechart` and "wraps" it into one series per period, aligned on a common relative axis.

Example — day-over-day revenue comparison

```
index=web action=purchase
| timechart span=1h sum(price) AS revenue
| timewrap 1d
```

The result has one column per day (`latest_day`, `1day_before`, ...) sharing a 24-hour x-axis — so a line chart lays each day on top of the others and seasonal patterns jump out.

Option	Effect
<code>timewrap 1w</code>	Compare weeks (aligned by weekday).
<code>series=short exact</code>	Control the generated series names.
<code>align=now end</code>	Anchor the wrap window.

GREAT FOR DASHBOARDS

A timewrapped panel is the classic "vs. last period" comparison — pair it with a single-value showing the percentage change to headline whether things are up or down.

10 · Anomaly Detection & Forecasting

A family of commands finds the unusual and projects the future — the on-ramp to analytics without a data-science stack. Most operate on a numeric series produced by `timechart`.

Command	What it does
<code>bin</code> (bucket)	Group numeric/time values into buckets — the foundation timechart is built on.
<code>trendline</code>	Moving averages (<code>sma</code> , <code>ema</code> , <code>wma</code>) to smooth a series and reveal trend.
<code>anomalydetection</code>	Flags outlier events/values by probability (supersedes <code>anomalousvalue</code> / <code>outlier</code>).
<code>predict</code>	Forecasts future values of a time series with confidence intervals.
<code>appendcols</code>	Appends the columns of a subsearch side-by-side (join by position) — used to align a forecast or trend next to actuals.

Example — smooth a series with a trendline

```
index=web action=purchase
| timechart span=1d sum(price) AS revenue
| trendline sma3(revenue) AS trend
```

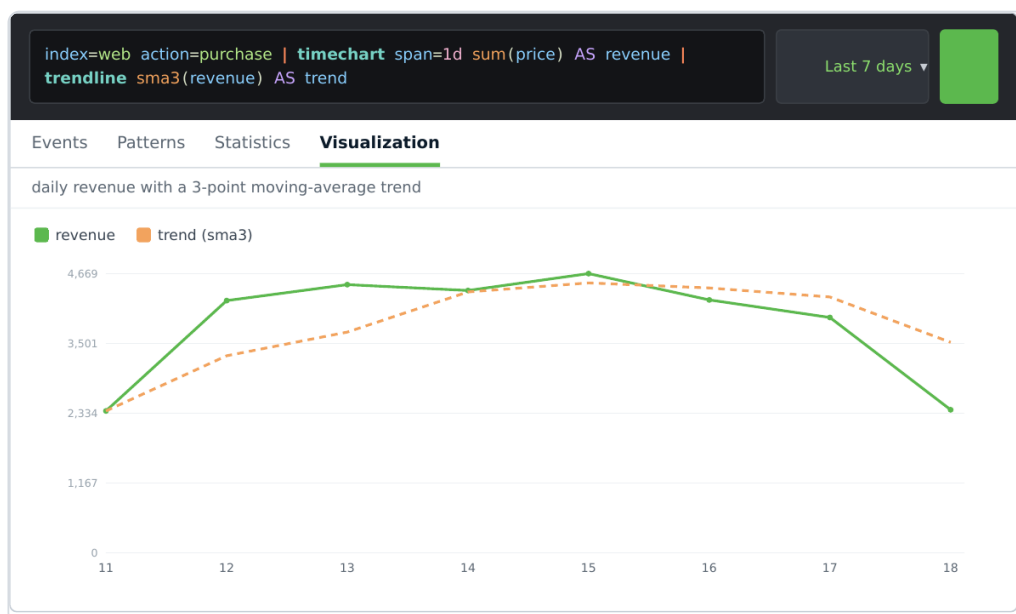


Fig 10.1 — Daily revenue (solid) with a 3-point moving-average trend (dashed) smoothing the noise.

Example — flag anomalous hours

```
index=security vendor_action="Failed password"
| timechart span=1h count AS fails
| anomalydetection fails
```

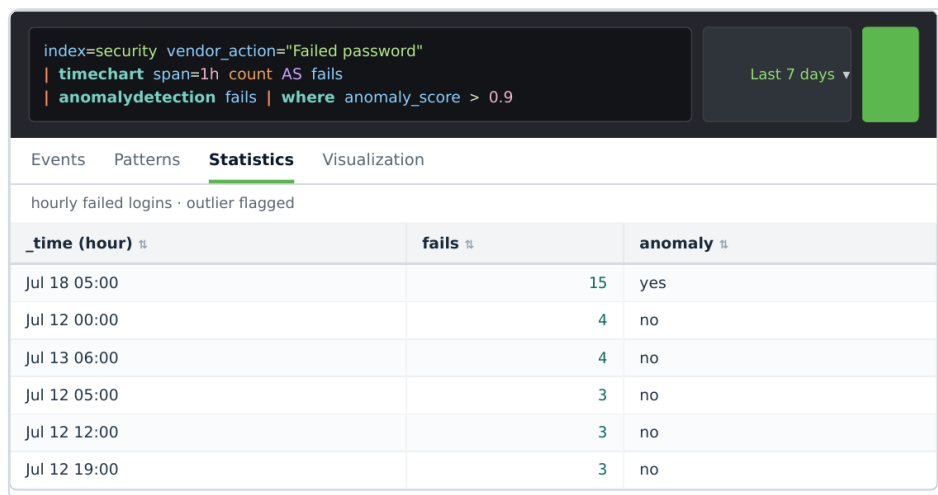


Fig 10.2 — The brute-force hour (15 fails) stands out as the anomaly against a quiet baseline.

Forecast the next points

```
| timechart span=1d sum(price) AS revenue | predict revenue future_timespan=3
```

DATA HUNGER

`predict` and robust anomaly detection want enough history to learn a pattern (ideally weeks/seasons). On a short window they still run, but treat the output as illustrative — more data, better model.

11 · Data Models & Pivot

Day 3 built and accelerated data models for `tstats`; the other half of a data model is **Pivot** — a no-SPL interface where analysts drag fields to build tables and charts against the model's curated schema. The `pivot` command is the SPL form of what the Pivot UI generates.

The two ways to consume a model

Path	Who & how
Pivot UI	Analysts drag "split rows/columns" and "column values" — Splunk writes the search.
<code> pivot</code>	SPL form: <code>pivot <model> <dataset> <agg> BY <fields></code> .
<code> tstats ... FROM datamodel=...</code>	Fastest, over acceleration summaries (Day 3).
<code> datamodel ...</code>	Inspect or run a dataset as a raw search.

Example — Pivot in SPL

```
| pivot Buttercup_Sales Purchases sum(price) AS revenue BY categoryId SPLITROW _time PERIOD=day
```

WHY MODELS PAY OFF

Curate the schema once (constraints, calculated fields, CIM names), and both analysts (Pivot) and power users (`tstats` / `pivot`) work fast against the same clean, accelerated definition — the payoff for the modelling effort from Day 3.

PART B

User Search Problems

"My search is wrong / slow / empty." Most problems fall into a handful of recognisable classes, and Splunk gives you the tools — the Job Inspector, job messages, and a methodical bisection approach — to isolate any of them quickly.

12 · Troubleshooting Searches

Debugging a search is a discipline: recognise the class of problem, then isolate the offending stage. The pipeline structure makes this tractable — you can inspect the data at every `|`.

Types of search problem

Symptom	Usual cause	First check
No results	Wrong index/time/field name; case mismatch	Widen time; remove filters one by one; verify field exists.
Wrong / partial results	Subsearch/join hit a cap; NULLs dropped; type mismatch	Job messages for "finalized"; check counts before/after the command.
Too slow	All-time range; late filtering; expensive command	Job Inspector execution costs; filter earlier.
Field missing	Extraction/alias/lookup didn't apply; wrong sourcetype	Search-time order; <code> fieldsummary</code> ; check props scope.
Wrong numbers	String math; double-counting; multivalue	Cast with <code>tonumber</code> ; inspect a single event.

Isolate methodically

1. **Bisect the pipeline** — delete commands from the end back toward the search until results look right; the last command you removed is the culprit.
2. **Read job messages** — yellow/red banners announce truncated subsearches, missing lookups, and quota limits.
3. **Open the Job Inspector** — total time, scan count, and per-command execution costs show *where* the time and rows go.
4. **Inspect one event** — verify a field's real name, case and type before blaming the aggregation.

Search Job Inspector — this search has completed		
Total time 8.42 s	Events matched 6,000	Scan count 512,904
Execution costs (top contributors)		
Component	Duration (s)	Invocation
command.search.index	0.62	
command.search.kv (extraction)	5.90	6,000
command.search.rawdata	1.41	
command.stats	0.31	
► field extraction dominates — an indexed field or tstats would cut most of this.		

Fig 12.1 — The Job Inspector: here field extraction (`command.search.kv`) dominates the 8.4 s — the thing to optimise.

READ THE COSTS LIKE A PROFILER

`command.search.index` = finding events; `command.search.rawdata` = reading them; `command.search.kv` = extracting fields; `command.<name>` = each SPL command. The biggest number tells you what to fix — usually "read/extract fewer events" (filter earlier) or "use tstats".

BEWARE SILENT TRUNCATION

The single most common "wrong results" bug is a subsearch or `join` quietly hitting its 10k/50k cap. If a correlation looks incomplete, check the job messages first — no error doesn't mean no problem.

PART C

Performance Tuning

Fast Splunk is mostly about doing less work: touch fewer events, extract fewer fields, and let acceleration do the heavy lifting. This part covers search-level optimization and the platform-level factors — indexing throughput and hardware — that set the ceiling.

13 · Making Splunk Fast — Search Optimization

Nearly every slow search is slow for the same reason: it reads too many events, too late. The fixes are consistent, and each maps directly to a number in the Job Inspector.

The optimization playbook

Technique	Why it helps
Narrow the time range	Time is the cheapest filter — Splunk skips whole buckets it never opens.
Filter in the base search	Index/terms/field filters happen before retrieval; keywords use the index directly.
Be specific: index, sourcetype	Avoid <code>index=*</code> ; name the index so Splunk searches only what it must.
Trim fields early (<code>fields</code>)	Less data flows to later commands; reduces extraction and transport.
Prefer <code>tstats</code>	Counts/aggregates from tsidx/DM summaries skip reading raw events entirely.
Replace join/subsearch with <code>stats</code>	Uncapped, streaming-friendly, no silent truncation.
Put streaming commands before centralised ones	Let indexers do <code>eval</code> / <code>where</code> / <code>rex</code> before the search head runs <code>stats</code> .

Optimization impact — same result, less work

Technique	Before	After
Filter early (index/time/terms)	<code>index=* search action=purchase</code>	<code>index=web action=purchase earliest=-24h</code>
Replace join with stats	<code>... join clientip [search ...]</code>	<code>... stats ... by clientip</code>
Use tstats over raw stats	<code>index=web stats count by action</code>	<code> tstats count where index=web by action</code>
Drop fields early	<code>... full events ...</code>	<code>... fields _time action price ...</code>

Fig 13.1 — Common rewrites and their impact: filter early, drop join, use tstats, trim fields.

tstats & data-model acceleration tuning

Lever	Guidance
<code>summariesonly=t</code>	Query only accelerated summaries — fastest; misses un-summarised recent data unless <code>allow_old_summaries=t</code> .
Accelerate the right range	Match the acceleration window to what you actually query; don't summarise years you never search.
Keep models lean	Fewer datasets/fields = smaller, faster-to-build tsidx summaries.
Schedule-friendly backfill	Let acceleration build during quiet hours; monitor completion under Settings → Data models.
<code>prestats=t</code> → chart	Feed <code>tstats</code> straight into <code>timechart</code> / <code>chart</code> for fast visuals.

MEASURE, THEN OPTIMISE

Change one thing, re-run, read the Job Inspector. Guessing wastes time; the execution costs tell you exactly which stage improved — and whether your change actually helped.

14 · Platform, Indexing & Resource Planning

Search speed has a ceiling set by the platform. Indexing throughput, storage speed and core count determine how much data Splunk can ingest and how many searches it can run concurrently.

Indexing throughput

Factor	Effect on ingest
Parsing & line breaking	Well-defined props (explicit <code>TIME_FORMAT</code> , <code>LINE_BREAKER</code> , <code>SHOULD_LINEMERGE=false</code>) cut CPU per event dramatically.
Index-time fields & transforms	Minimise index-time work; push extraction to search time unless a field is filtered constantly at scale.
Parallel pipelines	More ingestion pipelines raise throughput on multi-core indexers.
Forwarder load balancing	Spread data across indexers so no single node is a bottleneck.

Resource planning — the big three

Resource	Bounds	Plan for
CPU cores	Concurrent searches & ingestion pipelines	Peak concurrent searches (scheduled + ad-hoc); ~1 core per busy search.
IOPS / disk	The usual bottleneck — event retrieval is I/O bound	Fast local SSD/NVMe for hot/warm; Splunk's reference is high sustained IOPS.
Memory	Large <code>stats</code> / <code>transaction</code> /lookups, KV Store	Headroom for concurrent memory-heavy searches; KV Store working set.

Avoid the expensive patterns

Pattern	Problem	Do instead
<code>join</code>	50k cap, centralised, silent truncation	<code>stats ... BY key</code> / <code>lookup</code>
Subsearch	10k / 60s cap, sequential	Keep tiny, or use <code>stats</code>
<code>transaction</code>	Memory-heavy, non-streaming	<code>stats</code> with first/last/range
<code>index=*</code> / all-time	Scans everything	Name index; bound time
Late <code>eval</code> / <code>rex</code> on huge sets	Runs per event unnecessarily	Filter first; extract only what's kept

DESIGN FOR THE PLATFORM

The fastest search still runs on real hardware. Good props (clean parsing), acceleration where it pays, and disciplined SPL together let a well-sized cluster serve many users — the difference between a Splunk that scales and one that crawls.

15 • Appendix A — Expensive-Pattern → Fast-Alternative Guide

The single most valuable habit from Day 5: recognise a costly pattern and know its faster equivalent.

Instead of...	Prefer...	Why
... join key [search ...]	(A OR B) stats ... BY key	No 50k cap, streams
... [search ... top 1 x]	compute x with stats/eventstats	No 10k/60s cap
transaction key (for aggregates)	stats range(_time) count BY key	Streaming, low memory
index=web stats count	tstats count WHERE index=web	Reads tsidx, not raw
index=* ... all time	name index + bound time	Skips buckets
full events downstream	fields _time f1 f2 early	Less data to move
mvexpand then filter	mvfilter()/mvcount()	No event multiplication
lookup keyed on messy field	normalise key first (alias/eval)	Avoids silent misses

ONE RULE TO REMEMBER

Reduce the event set as early as possible, then prefer `stats` / `tstats` / `lookup` over `join` / `subsearch` / `transaction`. Almost every performance win on Day 5 is a version of this sentence.

16 · Appendix B — Day-5 Quick-Reference Cheat Sheet

CORRELATE & RESHAPE

```
| join type=left key [search ...] # last resort
(A OR B) | stats ... BY key # preferred
| transaction src_ip maxpause=5m
| xyseries row col val | untable row col val
```

WINDOWED & MULTIVALUE

```
| streamstats time_window=5m count BY src_ip
| eventstats sum(price) AS tot
| mvexpand pid mvcount() mvfilter()
split()
```

SUBSEARCH & TIME

```
... [search ... | return categoryId]
| timechart span=1h sum(price) | timewrap 1d
```

ANALYSE & FORECAST

```
| trendline sma3(revenue)
| anomalydetection fails
| predict revenue future_timespan=7
| pivot Buttercup_Sales Purchases sum(price)
BY categoryId
```

TROUBLESHOOT

```
# bisect: delete from the end back
# read job messages (truncation!)
# Job Inspector -> execution costs
# command.search.kv high? -> fewer events / tstats
```

GO FAST

Do	Not
name index + time	index=* all-time
tstats	stats on raw
stats/lookup	join/subsearch
fields early	carry all fields

END OF DAY 5 — AND THE TRACK

You can now correlate, reshape, expand and forecast with advanced SPL, troubleshoot any search with the Job Inspector, and tune both searches and the platform for speed at scale. Across five days you've gone from your first `index=web` to production-grade Splunk on one dataset — the Buttercup Games data in `Buttercup_Games_Dataset.zip`.